

Hardware Certifications & Environmental Standards: When organizations purchase hardware (procurement), they don't just look at speed; they look at the total lifecycle impact. This is where certifications come in.

EPEAT (Electronic Product Environmental Assessment Tool):

1. Definition: EPEAT is a global rating system used to evaluate the environmental impact of electronic products such as computers, laptops, and monitors.

2. Purpose: It helps organisations and buyers choose electronics that are environmentally friendly and support sustainable technology use.

3. Evaluation Criteria: EPEAT checks the entire product life cycle including material selection, reduction of toxic substances, product longevity, and ease of recycling at the end of life.

3. Tier System: EPEAT products are classified into three levels:

- **Bronze:** Meets all required environmental criteria.
- **Silver:** Meets all required criteria plus 50% of optional criteria.
- **Gold:** Meets all required criteria plus 75% of optional criteria.

4. Benefits: Using EPEAT-rated products helps reduce electronic waste, toxic materials, and environmental pollution.

5. Ex: A university purchasing 5,000 laptops may specify **EPEAT Gold** in the tender to ensure the devices are environmentally friendly and easier to recycle after their lifecycle.

RoHS (Restriction of Hazardous Substances):

1. Definition: RoHS is an international environmental regulation that restricts the use of hazardous substances in electrical and electronic products. It is often called the “lead-free directive.”

2. Origin and Purpose: The regulation started in the European Union and is now followed globally to ensure safer manufacturing of electronic devices.

3. Restricted Substances: RoHS limits harmful materials such as Lead (Pb), Mercury (Hg), Cadmium (Cd), Hexavalent Chromium, PBB, and PBDE, which can damage human health and the environment.

4. Importance: Before RoHS, electronic waste in landfills could release toxic substances like lead into soil and groundwater. RoHS encourages manufacturers to use safer materials such as lead-free solder.

5. Environmental Benefit: By reducing hazardous chemicals in electronics, RoHS helps decrease pollution and improves the recycling of electronic products.

6. Ex: To sell an iPhone in Europe, every component must be RoHS compliant. If even one part contains excessive hazardous material, the entire shipment can be rejected.

ENERGY STAR:

1. Definition: ENERGY STAR is an energy efficiency certification program developed by the U.S. Environmental Protection Agency (EPA) to promote energy-saving electronic products.

2. Purpose: The program helps consumers and organisations identify devices that use less electricity and reduce energy consumption.

3. Energy Efficiency Features: ENERGY STAR certified computers use efficient power supplies that convert electricity with minimal energy loss and produce less heat.

4. Power Management: These devices include power management features such as automatically entering sleep mode when they are idle, which helps save electricity.

5. Benefits: Using ENERGY STAR products reduces electricity costs, lowers carbon emissions, and improves overall energy efficiency.

6.Ex: In a data centre with 10,000 servers, using ENERGY STAR certified servers can reduce electricity consumption by about 20%, and less heat generation also reduces cooling costs.

Computers VS Servers:

Basis	Computers (Clients)	Servers
Purpose	Used by individual users for tasks like browsing, gaming, and office work.	Provide services and resources to multiple users or computers over a network.
Design Focus	Designed for user interaction and graphical performance.	Designed for reliability, stability, and continuous operation.
Performance Type	Provides burst performance for short, interactive tasks.	Built for handling many requests simultaneously (high concurrency).
Memory Type	Generally uses normal RAM.	Uses ECC RAM to detect and correct memory errors.
Reliability Features	Usually has single power supply and storage device.	Includes redundancy such as dual power supplies and RAID storage.
Uptime	Can be shut down or restarted frequently.	Designed to run continuously for long periods with minimal downtime.

Blade Servers:

1.Definition: Blade servers are a type of high-density server architecture designed to save space, reduce cabling, and improve efficiency in data centres.

2.Problem with Traditional Servers: Traditional rack servers are separate machines, each with its own power cable, cooling fans, and network connections, which creates complex cabling and management issues.

3.Blade Concept: A blade server is a compact server unit that contains essential components such as CPU, RAM, and storage and is inserted into a special frame called a chassis.

4.Blade Enclosure (Chassis): The chassis provides shared power supply, cooling systems, and networking for all blade servers, reducing hardware duplication and improving efficiency.

5.Advantages: Blade servers save physical space, reduce power consumption, simplify cabling, and make server management easier.

5.Ex: In a bank's transaction processing centre, instead of using 100 separate servers, organisations can use a few blade chassis that hold multiple blade servers, allowing many servers to run in a small space with fewer power cables.

Consolidation and Virtualization

1.Definition: Consolidation is the process of reducing the number of physical servers while still performing the same amount of computing work.

2.Server Sprawl Problem: Earlier, organisations used separate physical servers for different tasks such as email, file storage, and databases. Many of these servers used only a small portion of their CPU capacity, leading to wasted hardware, electricity, and space.

3.Virtualization Concept: Virtualization solves this problem by using one powerful physical server, called the **host**, to run multiple virtual systems.

4.Hypervisor: A special software called a **hypervisor** (such as VMware or Hyper-V) is installed on the host server to create and manage multiple **Virtual Machines (VMs)**.

5.Benefits: This approach reduces hardware requirements, saves electricity, lowers maintenance costs, and improves resource utilisation.

6.Ex: A logistics company may move servers from multiple branch offices into a single central cloud system. Employees then access their systems remotely, reducing the need for physical servers at each location.

Blade Servers and Virtualization:

1.Architecture: In blade server systems, computing and storage are separated. Blade servers provide processing power (CPU and RAM), while data is stored in a separate system called a SAN (Storage Area Network).

2.Blade Servers: Blade servers mainly act as processing units with powerful CPU and memory. They have little local storage because most data and operating systems are stored in the SAN.

3.Hypervisor: A Type-1 Hypervisor such as VMware ESXi or Microsoft Hyper-V is installed directly on the hardware. It manages the server and allows multiple Virtual Machines (VMs) to run on a single blade server.

4.Live Migration: Virtual machines can be moved from one blade server to another without stopping the system. This helps in maintenance and improves system flexibility.

5.High Availability (HA): If one blade server fails, the system automatically restarts the virtual machines on another server to avoid downtime.

6.Dynamic Resource Scheduling (DRS): The system monitors workload and automatically balances virtual machines between servers to maintain good performance and efficient resource use.

Remote Desktop & Connectivity: Remote Desktop allows a user to view and control the desktop of a computer from a remote location.

Remote Desktop Protocol (RDP):

1.Definition: Remote Desktop Protocol (RDP) is a technology that allows a user to access and control a computer from another location through a network or the internet.

2.Working Principle: In remote desktop, files are not transferred between computers. Instead, user inputs are sent and screen images are received.

2.Input Transmission: When a user moves the mouse or types on the keyboard, those actions are sent as input commands to the remote computer.

3.Output Transmission: The remote computer processes the input and sends back an updated image of the screen to the user's device.

4.Communication Process: The remote computer continuously sends compressed screen updates while receiving input commands from the user.

5.Result: This process allows users to control a distant computer as if they were physically using it.

Establishing a Remote Desktop Connection:

1.Connection Request: The client computer sends a connection request to the host computer using its IP address and a specific port (usually Port 3389 in RDP).

2.Negotiation: The client and host systems communicate to decide settings such as encryption level, screen resolution, and connection parameters.

3.Authentication: The user must verify their identity by entering username, password, or multi-factor authentication (MFA) to access the remote system.

4.Session Creation: After successful authentication, the host computer creates a remote desktop session for the user.

5.Virtual Display: The host system generates a virtual display driver that sends the screen output through the network instead of the physical monitor.

6.Remote Control: The user can now control the remote computer from their device as if they were sitting in front of it.

Real-Life Example: The IT Admin

Scenario: A server crash at 2:00 AM.

With Remote Desktop: The admin wakes up, opens a laptop, establishes a VPN for security.

With Remote Desktop Connection: Open RDP, enters the server's IP, and fixes the issue in 5 minutes from bed.

Remote Desktop Technology:

1.Definition: Remote Desktop technology allows users to access and control another computer from a different location through a network or internet connection.

2.Working Concept: It allows users to use the remote computer with full functionality, as if they were physically sitting in front of that computer.

3.Connection Requirements: To establish a remote connection, the user needs the IP address of the host computer, login credentials, and a stable internet connection.

4.Environmental Benefits: Remote desktop reduces the need for physical travel to offices, which saves fuel and lowers pollution, supporting green computing.

5.Productivity: Employees can work, access files, and run applications from anywhere while maintaining collaboration and efficiency.

6.Ex: An employee working from home can connect to their office computer using remote desktop and perform their tasks remotely.

Greening Your Information Systems: This focuses on the practical steps and physical changes a business can make to reduce the environmental impact of its technology and operations.

Initial Improvement Calculations:

1.Meaning: Initial improvement calculations refer to measuring the current environmental and energy impact of IT systems before implementing Green IT solutions.

2.Purpose: The main purpose is to understand the present level of energy consumption, carbon emissions, and operational costs in an organisation.

3.Data Analysis: This process involves analysing electricity bills, hardware power ratings, server usage, and other resource consumption to establish a baseline.

4.Estimating Improvements: After collecting the data, organisations estimate how much energy, cost, and environmental impact can be reduced by adopting greener technologies.

5.Ex: For instance, a company running 50 physical servers spends \$10,000 annually on electricity. By virtualizing them into 5 servers, they estimate saving about \$8,000 per year while reducing energy consumption.

Selecting Metrics:

1.Meaning: Selecting metrics means choosing specific measurement units to evaluate and monitor the environmental performance of IT systems.

2.Importance: It is important because organisations cannot effectively manage or improve sustainability without measuring their resource usage.

3.Common Metrics: Typical Green IT metrics include kilowatt-hours (kWh) for energy consumption, Power Usage Effectiveness (PUE) for data centre efficiency, and paper usage for office operations.

4.Purpose: These metrics help organisations track progress, identify inefficiencies, and measure improvements in energy and resource usage.

5.Ex: A data centre may choose PUE as its main metric. A PUE of 2.0 means that for every watt used by computing equipment, another watt is used for cooling and overhead. The goal is to reduce it to 1.2 to improve efficiency.

Tracking Progress:

1.Meaning: Tracking progress means regularly monitoring the selected green metrics to evaluate whether sustainability initiatives are effective.

2.Purpose: It helps organisations understand if their actions are reducing energy consumption, resource usage, and environmental impact.

3.Continuous Monitoring: The data is collected and analysed over time to identify improvements or areas where further changes are needed.

4.Performance Evaluation: By comparing current results with previous data, organisations can measure the success of their Green IT policies.

5.Ex: An IT manager may create a monthly dashboard showing the office's electricity consumption to check if the policy of turning off computers at night is reducing energy usage.

Change Business Processes:

1.Meaning: Changing business processes means modifying daily work activities and workflows to make them more environmentally friendly.

2.Purpose: The goal is to reduce energy use, waste, and carbon emissions by improving how work is performed, not just by using greener technology.

3.Process Improvement: Organisations review their existing procedures and identify areas where resources such as paper, electricity, and travel can be reduced.

4.Sustainable Practices: Digital communication, online documentation, and remote collaboration tools can replace traditional resource-intensive processes.

5.Ex: A company replaces in-person HR onboarding that required travel with online training through video meetings and digital modules, reducing carbon emissions and costs.

Customer Interaction:

1.Meaning: Customer interaction in Green IT means encouraging customers to participate in environmentally friendly initiatives.

2.Purpose: It helps extend the organisation's environmental efforts beyond the company by involving customers in sustainable practices.

3.Green Practices: Companies promote activities such as paperless billing, digital receipts, and online services to reduce paper usage and waste.

4.Environmental Impact: When customers adopt these practices, it reduces resource consumption and supports overall sustainability goals.

5.Ex: A telecom company may offer a \$2 monthly discount to customers who choose paperless e-billing instead of receiving printed bills by mail.

Paper Reduction:

1.Meaning: Paper reduction refers to the effort to minimise or eliminate the use of physical paper in offices by using digital alternatives.

2.Purpose: The aim is to save natural resources such as trees and water that are used in paper manufacturing.

3.Digital Alternatives: Organisations can use electronic documents, emails, cloud storage, and digital signatures instead of printed documents.

4.Benefits: Reducing paper usage also saves office storage space, lowers printing costs, and decreases environmental impact.

5.Ex: A law firm may replace physical contract signing with digital signature software such as DocuSign, saving thousands of printed pages every month.

Green Supply Chain:

1.Meaning: Green supply chain refers to selecting suppliers and partners who follow environmentally friendly practices.

2.Purpose: The aim is to ensure that the entire lifecycle of a product, from production to delivery, is sustainable and eco-friendly.

3.Sustainable Procurement: Companies choose vendors that use renewable energy, reduce waste, and follow environmental regulations.

4.Environmental Impact: A green supply chain helps reduce carbon emissions, resource consumption, and pollution across the production process.

5.Ex: Apple requires its manufacturing partners to use 100% renewable energy for producing Apple devices. Suppliers who do not follow this requirement may lose their contract.

Improve Technology Infrastructure:

1.Meaning: Improving technology infrastructure means upgrading existing IT systems such as hardware, storage, and networks to more energy-efficient technologies.

2.Purpose: The goal is to reduce power consumption, improve system performance, and support environmentally friendly computing.

3.Modern Technologies: Organisations can adopt efficient hardware, advanced storage systems, and optimized network equipment that use less electricity.

4.Benefits: Energy-efficient infrastructure reduces electricity costs, produces less heat, and improves overall system speed and reliability.

5.Ex: An office may replace traditional Hard Disk Drives (HDDs) with Solid State Drives (SSDs), which consume less power, generate less heat, and provide faster performance.

Reduce PCs and Servers:

1.Meaning: Reducing PCs and servers means decreasing the number of physical machines used in an organisation to improve efficiency and save energy.

2.Purpose: The aim is to lower electricity consumption, reduce hardware costs, and minimise environmental impact.

3.Hardware Consolidation: Organisations combine multiple computing tasks into fewer machines so that resources are used more efficiently.

4.Virtualization: This is commonly achieved through virtualization, where multiple virtual servers run on a single powerful physical server.

5.Ex: Instead of using separate servers for email, databases, and intranet services, a company can run all these systems as virtual machines on one server, reducing hardware and energy usage.

Shared Services:

1.Meaning: Shared services refer to the practice of centralising resources so that multiple departments or users share the same infrastructure.

2.Purpose: The aim is to reduce the duplication of resources, lower operational costs, and improve efficiency.

3.Centralised Infrastructure: Instead of each department maintaining its own systems, services such as storage, servers, and applications are provided from a central system.

4.Cloud Computing: Cloud computing is a common example of shared services where many users access the same computing resources through the internet.

5.Ex: Instead of providing personal printers for 50 employees, a company installs two shared multi-function printers that everyone can use, reducing energy use and equipment costs.

Hardware Costs:

1.Meaning: Hardware cost analysis in Green IT focuses on evaluating the financial impact of purchasing energy-efficient technology.

2.Initial Investment: Green hardware often has a higher upfront purchase price compared to standard hardware.

3.Operational Savings: Energy-efficient devices consume less electricity and produce less heat, which reduces power and cooling costs.

4.Long-Term Benefit: Over time, the lower operational expenses can offset the initial cost and provide financial savings.

5.Ex: An Energy Star certified monitor may cost \$20 more than a normal monitor, but it can save about \$30 in electricity during its lifespan, resulting in overall cost savings.

Cooling:

1.Meaning: Cooling in Green IT refers to managing the heat produced by computers and servers in an energy-efficient way.

2.Need for Cooling: IT equipment such as servers generates a large amount of heat, and traditional cooling methods like air conditioning consume a lot of electricity.

3.Green Cooling Techniques: Green computing encourages efficient thermal management methods to reduce energy use while maintaining proper system temperature.

4.Benefits: Efficient cooling reduces electricity consumption, lowers operational costs, and improves the lifespan of IT equipment.

5.Ex: A data centre may use a Hot Aisle/Cold Aisle arrangement, where hot air from servers is separated from cold air, allowing cooling systems to work more efficiently and use less energy.

Staying Green: This is about sustainability over the long term. It focuses on corporate culture, management, and continuous improvement so the company doesn't slip back into old, wasteful habits.

Organizational Check-ups:

1.Meaning: Organizational check-ups are regular reviews of an organisation's environmental policies and green practices.

2.Purpose: The aim is to check whether sustainability policies are being properly followed and are still effective.

3.Monitoring: These reviews help organisations track energy usage, recycling activities, and electronic waste management.

4.Improvement: Based on the results, organisations can improve policies, correct mistakes, and introduce better green practices.

5.Ex: Every six months, a company committee may review recycling rates, energy consumption, and e-waste records to ensure departments are following green policies.

Chief Green Officer (CGO):

1.Meaning: A Chief Green Officer (CGO) is a senior executive responsible for managing and promoting environmental sustainability within an organisation.

2.Role: The CGO develops and implements green strategies to reduce environmental impact and promote sustainable practices.

3.Responsibilities: They oversee initiatives such as energy efficiency, waste reduction, renewable energy use, and environmentally friendly business operations.

4.Policy Implementation: The CGO ensures that the organisation follows environmental policies and meets sustainability goals.

5.Ex: A large technology company may appoint a CGO to manage its transition to 100% solar power and ensure that new data centres follow green building standards.

Evolution in Green IT:

1.Meaning: Evolution in Green IT refers to the continuous improvement and development of environmentally sustainable technologies and practices.

2.Ongoing Process: Green IT is not a one-time initiative; organisations must regularly update their strategies as new technologies and environmental knowledge develop.

3.Adapting to Technology: Companies adopt advanced solutions such as energy-efficient systems, automation, and smart technologies to improve sustainability.

4.Environmental Responsibility: Continuous improvement helps organisations reduce carbon emissions, conserve resources, and meet changing environmental standards.

5.Ex: Earlier, companies focused on simple practices like paper recycling. Today, they use technologies such as AI to optimise building temperatures and invest in carbon capture solutions to reduce environmental impact.

Sell the CEO:

1.Meaning: "Sell the CEO" means convincing top management to support and approve Green IT initiatives.

2.Need for Approval: Many environmental projects require initial investment, so managers must explain their value to company executives.

3.Focus on ROI: IT managers present financial benefits such as cost savings, efficiency improvements, and return on investment (ROI).

4.Business Justification: By showing both economic and environmental advantages, executives are more likely to approve green technology projects.

5.Ex: An IT Director may show the CEO that installing smart HVAC systems costs \$50,000 but saves \$20,000 per year in electricity, recovering the cost in about 2.5 years.

SMART Goals:

1.Meaning: SMART goals are objectives that are **Specific, Measurable, Achievable, Relevant, and Time-bound**, which makes them clear and practical.

2.Purpose: They help organisations set clear environmental targets and track progress effectively.

3.Specific and Measurable: Goals should clearly define what needs to be achieved and include measurable results.

4.Achievable and Relevant: The goals should be realistic and aligned with the organisation's sustainability strategy.

5.Time-bound: A deadline should be set so that progress can be monitored within a specific time period.

6.Ex: A vague goal like *"We will use less power"* can be replaced with a SMART goal such as *"Reduce data centre electricity consumption by 15% by the end of 2026 through server virtualization."*

Equipment Check-ups:

1.Meaning: Equipment check-ups refer to regular maintenance of IT hardware to ensure it operates efficiently and safely.

2.Purpose: The aim is to keep computers, servers, and other devices working at peak performance while reducing unnecessary power consumption.

3.Preventive Maintenance: Regular inspection, cleaning, and servicing of equipment help prevent hardware problems and improve system lifespan.

4.Energy Efficiency: Well-maintained hardware uses less electricity because it does not need to work harder to maintain performance.

5.Ex: An IT technician may clean dust from server vents. If vents are blocked, cooling fans run at high speed, consuming more power and increasing the risk of overheating.

Gather Data:

1.Meaning: Gathering data means collecting raw information about how resources such as electricity, paper, and hardware are being used in an organisation.

2.Purpose: This step helps organisations understand their current resource consumption and environmental impact.

3.Data Collection: Information is collected from devices, electricity meters, hardware usage reports, and monitoring tools.

4.Importance: Accurate data collection is necessary to identify areas where energy and resources are being wasted.

5.Ex: Smart plugs can be installed on office devices like printers or refrigerators to measure how much electricity they use during a 24-hour period.

Tracking the Data:

1.Meaning: Tracking the data means recording the collected information over time using software systems or logs.

2.Purpose: It helps organisations observe patterns and trends in resource usage.

3.Monitoring: Regular tracking allows managers to compare daily, weekly, or monthly usage levels.

4.Decision Support: This information helps in making better decisions to reduce energy consumption.

5.Ex: Data collected from smart plugs can be displayed on a central dashboard that shows daily electricity usage through graphs.

Baseline Data:

1.Meaning: Baseline data refers to the initial measurements taken before any Green IT improvements are implemented.

2.Purpose: It provides a starting point that allows organisations to measure future improvements.

3.Importance: Without baseline data, it is difficult to prove that sustainability initiatives have reduced resource usage.

4.Comparison: Future performance is compared with this initial data to evaluate progress.

5.Ex: An office may record that it uses 500 reams of paper per month before introducing digital devices to reduce paper consumption.

Benchmarking:

1.Meaning: Benchmarking involves comparing an organisation's environmental performance with industry standards or competitors.

2.Purpose: It helps organisations understand whether they are performing better or worse than others in the same industry.

3.Performance Evaluation: Companies use benchmarking to identify areas where improvements are needed.

4.Standards Comparison: Industry reports and environmental guidelines are often used for comparison.

5.Ex: An airline may compare its data centre energy efficiency with EPA industry averages to evaluate its performance.

Analyse Data:

1.Meaning: Data analysis involves carefully examining collected data to identify inefficiencies and opportunities for improvement.

2.Purpose: The goal is to find patterns, unusual activity, or areas where resources are being wasted.

3.Decision Making: Analysis helps managers make informed decisions about improving systems and processes.

4.Problem Identification: It can reveal issues such as high power usage or inefficient operations.

5.Ex: An IT manager analysing electricity data may notice a spike at 2:00 AM due to automatic backups and fix it by scheduling backups at different times.

Conduct Audits:

1.Meaning: Conducting audits means carrying out a formal review of an organisation's environmental practices and data.

2.Purpose: The aim is to verify whether the organisation's green policies and sustainability claims are accurate and properly implemented.

3.External Verification: Audits are often performed by independent or external auditors to ensure transparency and credibility.

4.Compliance Check: They help organisations confirm that they are following environmental standards, regulations, and internal green policies.

5.Ex: An independent auditor may verify that a company is truly carbon neutral before the company is allowed to display that claim on its products.

Certifications:

1.Meaning: Certifications are official recognitions given to organisations that meet specific environmental standards.

2.Purpose: They show that a company follows sustainable practices and meets recognised environmental guidelines.

3.Credibility: Certifications help improve a company's reputation and build trust among customers and stakeholders.

4.Standards: These certifications are usually awarded by recognised organisations after careful evaluation.

5.Ex: A company may receive LEED (Leadership in Energy and Environmental Design) certification for constructing a building that uses natural lighting and recycled materials.

Benefits:

1.Meaning: Benefits refer to the positive outcomes of implementing Green IT practices in an organisation.

2.Financial Benefits: Green IT helps reduce electricity costs, maintenance expenses, and operational costs.

3.Environmental Benefits: It helps lower carbon emissions, reduce waste, and conserve natural resources.

4.Social Benefits: Sustainable practices improve company reputation and can increase employee satisfaction.

5.Ex: Remote work reduces office electricity use and commuting, leading to lower costs and happier employees.

Realities:

1.Meaning: Realities refer to the practical challenges organisations face while implementing Green IT initiatives.

2.High Initial Costs: Many green technologies require a large initial investment.

3.Resistance to Change: Employees and management may resist changing traditional working methods.

4.Technological Limitations: Some technologies still cannot completely eliminate emissions.

5.Ex: Installing a solar power system for an office may take 7–10 years to recover the high installation cost.

Helpful Organizations:

1.Meaning: Helpful organisations are groups, NGOs, or government bodies that support companies in implementing sustainable practices.

2.Guidelines and Frameworks: These organisations provide environmental standards, guidelines, and improvement frameworks.

3.Support and Resources: They may offer research, tools, training, or financial incentives for green initiatives.

4.Awareness and Education: They help spread knowledge about sustainability and encourage responsible technology use.

5.Ex: Organisations such as The Green Grid and the EPA (Environmental Protection Agency) provide frameworks and programs like Energy Star to help businesses improve energy efficiency.

Case Study & Surveys: Education for Sustainable Development in Computer Science:

1.Meaning: Education for Sustainable Development (ESD) in computer science focuses on teaching students how technology can be designed and used in an environmentally responsible way.

2.Purpose: The goal is to ensure that future software developers and IT engineers understand the environmental impact of technology.

3.Green Coding: Students are taught to write efficient and optimized code that uses less processing power and energy.

4.Sustainable System Design: The curriculum encourages designing software and systems that reduce resource consumption and support sustainable computing.

5.Ex: A university adds a course on algorithm efficiency, where students learn that poorly written code increases CPU usage and energy consumption, while optimized code saves energy and improves performance.